Mauro Patimo

PhD Student, Aerospace Engineering · The Pennsylvania State University

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RESEARCH INTERESTS

Gravity offloading and ground-based spacecraft testing; hardware-in-the-loop emulation of orbital motion, proximity operations, and in-space servicing, assembly, and manufacturing (ISAM); guidance, navigation, and control.

EDUCATION

PhD, Aerospace Engineering · The Pennsylvania State University — Expected May 2028

Advisor: Dr. Puneet Singla. Dissertation (in progress): *A 6DOF Gravity Offloading System*

M.S., Aerospace Engineering · The Pennsylvania State University — 2025

Advisor: Dr. Puneet Singla. Thesis: *A Modular Gravity Off-Loading System*

B.S., Astrophysics · California State University, Northridge — May 2023

Research advisor: Dr. Luca Ricci

RESEARCH EXPERIENCE

Graduate Researcher, Vehicle Dynamics & Controls · The Pennsylvania State University — May 2024 – Present

- Designing and building GRIP, a 9 m × 4 m × 3 m six-degree-of-freedom gravity offloading facility for satellite and ISAM testing (advisors: Dr. Puneet Singla, Dr. Roshan Eapen).
- Sized the X, Y, and Z axes against motor torque, reflected inertia, gearbox, rack, and bearing limits; validated the structure with finite-element analysis.
- Developed scaling criteria that map LEO, MEO, and GEO conditions onto the laboratory apparatus.
- Demonstrated circular-trajectory tracking with 0.35 mm mean radial error.
- M.S.: built a modular 5-DOF prototype (gantry and three-axis gimbal) that reproduced Keplerian, cislunar Lyapunov, and docking trajectories with sub-millimeter precision.

Research Intern, Plasma and Laser Physics · Princeton University — Jun 2022 – Aug 2022

Three-dimensional imaging with a femtosecond laser and ultra-high-speed liquid varifocal lenses (Dr. Craig Arnold).

Undergraduate Researcher, Biophysics · California State University, Northridge — Jan 2022 – Jul 2022

Image- and data-analysis software for Dr. Taheri, Department of Physics and Astronomy.

Undergraduate Researcher, Astrophysics · California State University, Northridge — Sep 2021 – Jun 2023

Data analysis of protoplanetary disks (Dr. Luca Ricci).

Undergraduate Researcher, Astrophysics · Los Angeles Pierce College — Jan 2020 – Jun 2021

Prediction of neutron star rotational speeds (Dr. Dale Fields).

TEACHING AND OUTREACH

Graduate Teaching Assistant · The Pennsylvania State University — Aug 2023 – May 2024
Office hours, examination grading, and solution keys.

Academic Tutor and Teaching Assistant · California State University, Northridge — Sep 2017 – May 2023
Tutored more than 400 students, over 90% of whom achieved a final grade of A.

Science Communication · YouTube — Jan 2020 – Present
Educational channel presenting physics and programming concepts.

TECHNICAL SKILLS

Mechanical design	Gantry and gimbal design, actuator sizing, FEA (Autodesk Fusion, FreeCAD)
Motion control	Teknic ClearPath servos, Raspberry Pi 5, multi-threaded C++, encoder data acquisition
Simulation	Trajectory and discretization-error simulation (Rust), orbital-condition scaling
Astrodynamics	Keplerian and elliptical orbits, cislunar Lyapunov orbits, docking emulation
Fabrication	Additive manufacturing (Prusa XL, PrusaSlicer), aluminum T-slot structures

HONORS AND AWARDS

- Sigma Pi Sigma, National Physics Honor Society
- 6th place, Kangourou della Matematica (national mathematics competition, Italy)
- 13th place, Olimpiadi Italiane di Astronomia (national astronomy competition, Italy)